In [16]:
```
import numpy as np
array=np.random.randint(20,100,20)
array
```

Out[16]:
```
array([44, 52, 47, 47, 69, 74, 79, 84, 98, 25, 40, 97, 35, 72, 64, 77, 61,
       91, 77, 84], dtype=int32)
```

In [17]:
```
array.mean()
```

Out[17]:
```
np.float64(65.85)
```

In [18]:
```
np.percentile(array,25)
```

Out[18]:
```
np.float64(47.0)
```

In [19]:
```
np.percentile(array,50)
```

Out[19]:
```
np.float64(70.5)
```

In [20]:
```
np.percentile(array,75)
```

Out[20]:
```
np.float64(80.25)
```

In [21]:
```
np.percentile(array,100)
```

Out[21]:
```
np.float64(98.0)
```

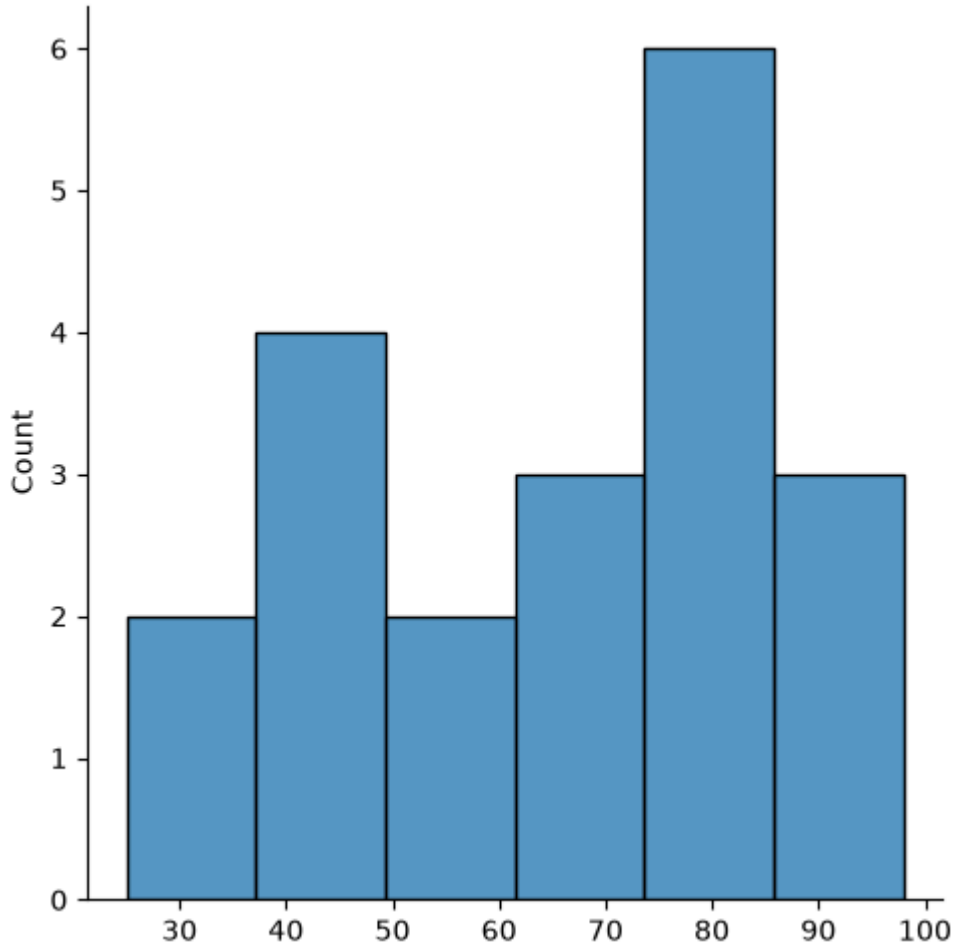
In [22]:
```
def outDetection(array):
    sorted(array)
    Q1,Q3=np.percentile(array,[25,75])
    IQR=Q3-Q1
    lr=Q1-(1.5*IQR)
    ur=Q3+(1.5*IQR)
    return lr,ur
```

In [23]:
```
lr,ur=outDetection(array)
lr,ur
```

Out[23]:
```
(np.float64(-2.875), np.float64(130.125))
```

In [24]:
```
import seaborn as sns
sns.displot(array)
```

Out[24]:
```
<seaborn.axisgrid.FacetGrid at 0x239d2b3f9d0>
```

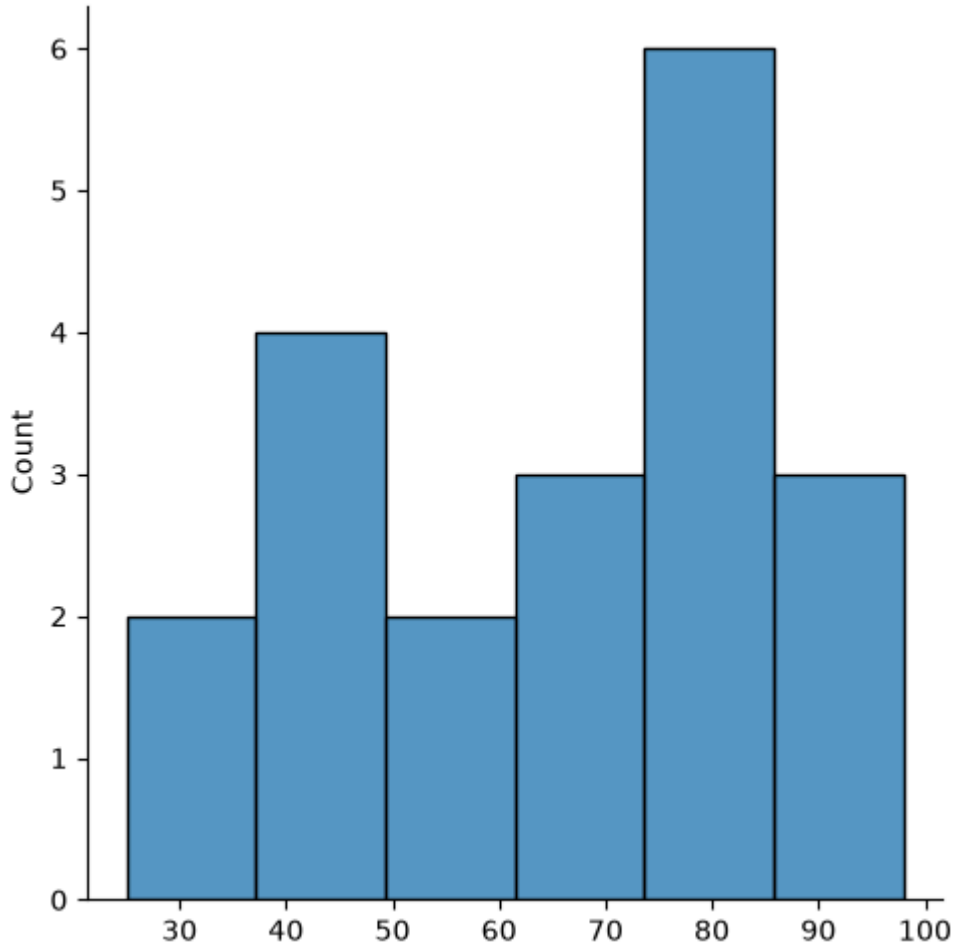


In [25]:
```
new_array=array[(array>lr) & (array < ur)]
new_array
```

Out[25]:
```
array([44, 52, 47, 47, 69, 74, 79, 84, 98, 25, 40, 97, 35, 72, 64, 77, 61,
       91, 77, 84], dtype=int32)
```

In [26]:
```
sns.displot(new_array)
```

Out[26]:
```
<seaborn.axisgrid.FacetGrid at 0x239d2bd6210>
```



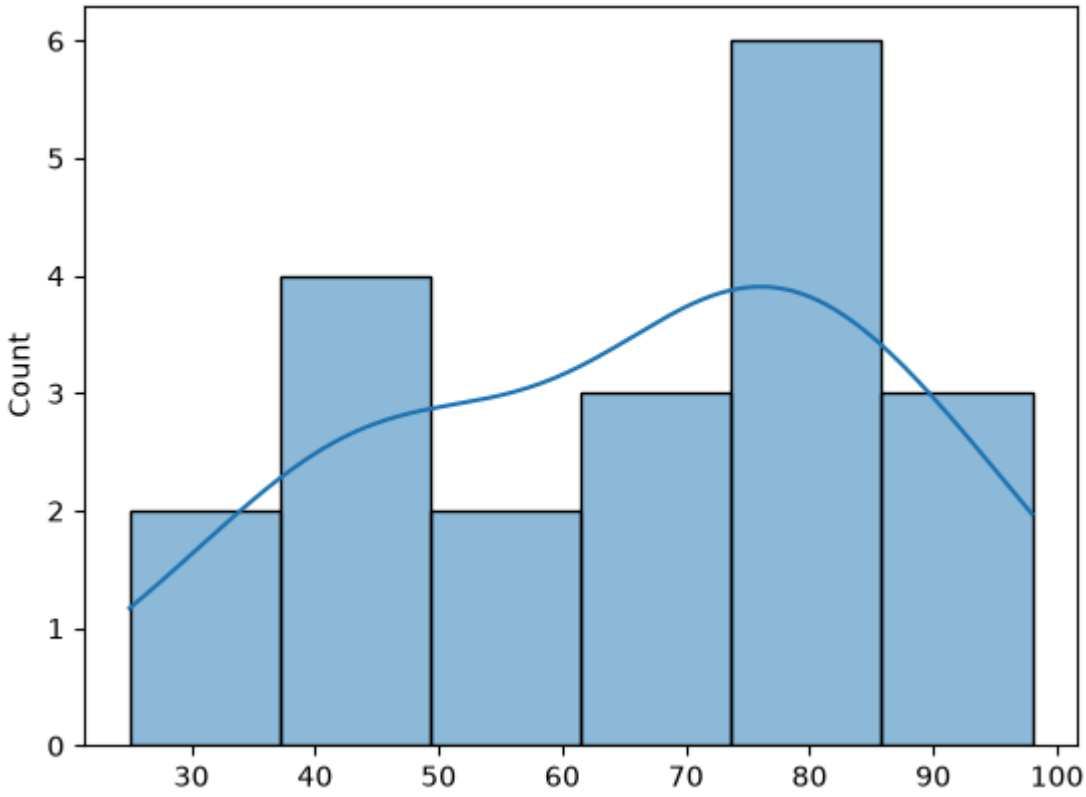
In [27]:
```
lr1,ur1=outDetection(new_array)
lr1,ur1
```

Out[27]:
```
(np.float64(-2.875), np.float64(130.125))
```

In [28]:
```
final_array=new_array[(new_array > lr1) &(new_array <ur1)]
final_array
```

Out[28]:
```
array([44, 52, 47, 47, 69, 74, 79, 84, 98, 25, 40, 97, 35, 72, 64, 77, 61,
       91, 77, 84], dtype=int32)
```

In [31]:
```
import seaborn as sns
import matplotlib.pyplot as plt
sns.histplot(final_array,kde=True)
plt.show()
```



In [ ]: